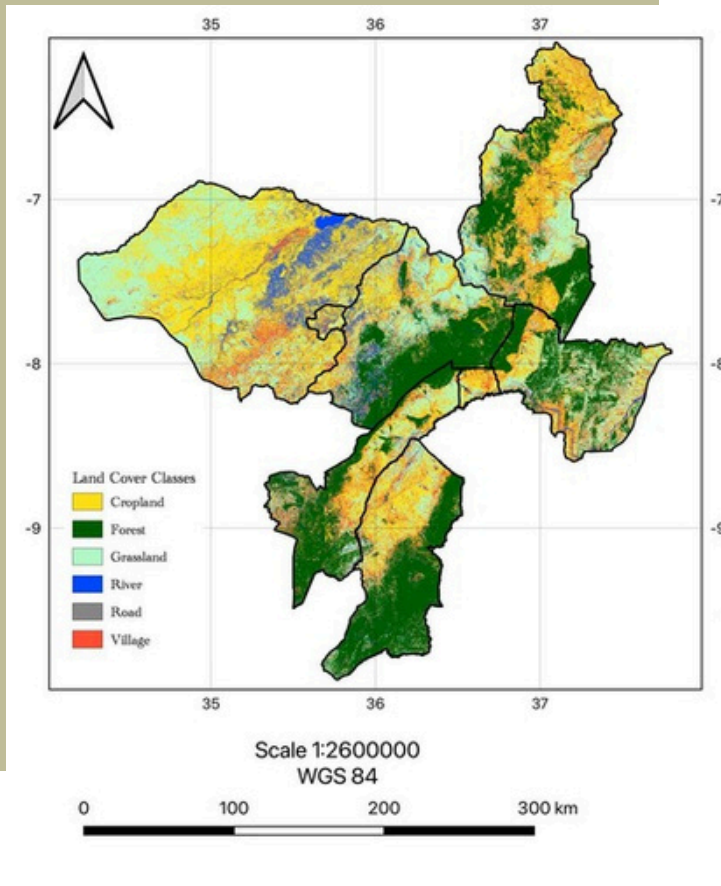




SUSTAIN-AFRICA **LAND HEALTH**

GUIDANCE BROCHURE

IUCN SUSTAIN-Africa



SUSTAIN Pro focuses on how to improve the sustainability of food systems and healthy productive landscapes. It has three main outcomes: (1) upscale solutions for sustainable agriculture, (2) restore land health through multi-stakeholder partnerships and (3) drive systemic change through investing in sustainable and inclusive value chains.

The Sustainability and Inclusion Strategy for Growth Corridors in Africa (SUSTAIN-Africa) is currently in phase II which is being delivered through SUSTAIN Productive Landscapes for Inclusive Growth (SUSTAIN Pro) and SUSTAIN Ecosystem stewardship to balance sustainability and growth (SUSTAIN Eco).

SUSTAIN Eco focuses on healthy ecosystems and prosperous communities in Tanzania, championing improved governance and rights, sustainable management and catalysing investment for the conservation and restoration of ecosystems and biodiversity.

SUSTAIN Land Health Monitoring Index



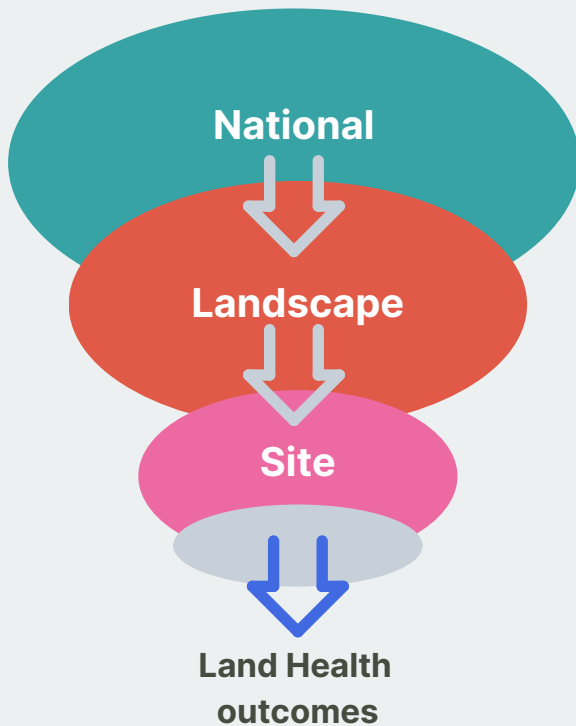
Land Health:
‘the capacity of land,
relative to its potential,
to sustain delivery of
essential ecosystem
services’ (Shepherd et
al., 2015).

For SUSTAIN, Land Health is pivotal in achieving scalability of nature-based solution interventions that improve sustainable agriculture, food systems and food security.

Measuring Land Health is vital for progression towards more sustainable systems because it facilitates understanding the current impact of pressures on soils/biodiversity/ecosystems, and provides evidence on what interventions improve or reduce the status of land health. This is important to know for monitoring of interventions and when scaling to other regions.

SUSTAIN needs to understand Land Health to link and track local-level interventions to national commitments and multilateral environmental agreements that will help catalyse investments in Nature-based solutions and system change.

SUSTAIN Land Health Monitoring Index



Pillars

The SUSTAIN Land Health Monitoring Index (LHMI) is built on three pillars: reliable and sufficient water of appropriate quality, soil health and resilience. These pillars interact across the three scales of site level, landscape level and national level.

Soil

Soil health is the variety of life below and above-ground at the site and farm level. It also includes the interactions of soil with water and climate, offering a whole view of soil health, such as soil moisture levels and carbon sequestration

Water

Reliable and sufficient water of appropriate quality reflects the need to ensure sustainable water use and well-maintained and managed water infrastructure that supports needed soil moisture in agricultural systems

Resilience

Resilience reflects how social-ecological systems need to be able to cope with stress and shocks and sustain the transformations needed to adapt to climate change while maintaining human wellbeing and economic production

Monitoring indicators

Indicator	Description
Site: Soil	
Bird species richness per site	A continuous indicator on the number of species per site
Invertebrate species richness per farm (pollinators and ground dwelling)	Measure of abundance and diversity of pollinators and ground dwelling invertebrates
Earthworm count (individuals and type)	Counts the number of each type of earthworm found at the plot level
Total Carbon stock soil and biomass (above and below ground)	An estimate of how much carbon is in above-ground biomass such as trees and crops and below ground in the soil
Soil C:N ratio	This is a measure of the concentration of total carbon to total nitrogen in the soil
soil pH	Acidity of soil
Rate of soil loss	Mean absolute change in pin height at each plot to indicate erosion/deposition
Site: Water	
Infiltration rate of water	This simple method measures the rate at which water can enter the soil
Depth to water table	Measures the depth in metres to the water table, the boundary of zone of aeration and zone of saturation in the soil
Diversity of aquatic macroinvertebrates	Using the Tanzania River Scoring System TARISSfupi.
Site: Water	
Access to water per different use types	Site level: assess farm water delivery systems Landscape level: Understand % access to water for different uses at the landscape scale: cooking, washing, cleaning, livestock, crop production
Diversity of aquatic macroinvertebrates	Using the Tanzania River Scoring System TARISSfupi.
Landscape: Resilience	
Reduction in household poverty	Uses the Poverty Probability Index (PPI) and is determined by establishing the percentage likelihood of the targeted population living below the poverty line of \$1.90 per person per day.
Natural ecosystem connectivity	Spatial analysis of the fragmentation of natural ecosystem patterns such as wildlife corridors and river catchment areas between clusters
Sustainable Land Management	A quantitative indicator to assess the adoption of different sustainable land management practices
Threatened species	A qualitative indicator in which the assessment team decides the field metrics
Extreme weather	Extreme weather (drought, flooding) is produced in regular reports by the Tanzanian meteorological society at the regional level
Agricultural, agroforestry and tree plantation productivity	A set of quantitative indicators to measure agricultural, agroforestry, and tree plantation productivity

Site: Soil

01

Bird species richness per site



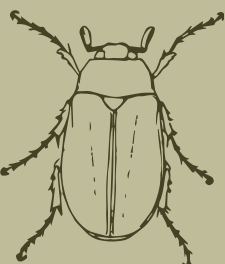
A continuous indicator on the number of species per site.

Ecosystem services/disservice: predators, pollinators, scavengers, seed dispersers, seed predators, and ecosystem engineers

Collection method: Field data collection yearly or every 3 years

02

Invertebrate richness per site



A continuous indicator on the number of ground dwelling and pollinator species per site.

Ecosystem services/disservices: pollination, pest control, nutrient cycling and regulation of water

Collection method: Field data collection yearly or every 3 years

03

Earthworm count



Count of earthworms and type at each site.

Ecosystem services/disservices: soil quality, bioaccumulation of toxic metals

Collection method: Field data collection yearly or every 3 years

04

Total Carbon Stocks



Total carbon in the biomass and soil

Ecosystem services/disservices: climate change mitigation, soil quality, water infiltration, flood mitigation

Collection method: Field data collection yearly or every 3 years

Site: Soil

05 Soil C:N ratio

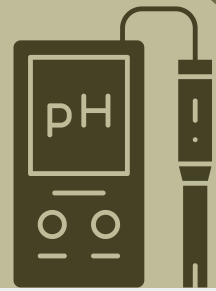


Assesses the relative proportion of carbon to nitrogen in soil

Ecosystem services/disservices: nutrient cycling and decomposition

Collection method: Field data collection yearly or every 3 years

06 Soil pH



Key chemical property

Ecosystem services/disservices: breakdown of organic matter, ammonia volatilisation, nitrification/denitrification, and toxic heavy metal concentrations.

Collection method: Field data collection yearly or every 3 years

07 Rate of soil loss



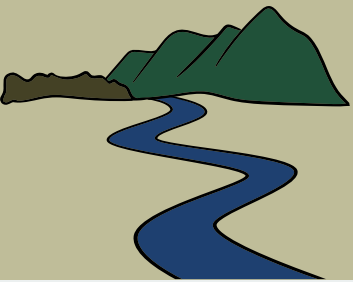
Soil erosion determines the rate at which soil is lost from a given area

Ecosystem services/disservices: fertility and productivity, yields and food security, water quality, sediment build up

Collection method: Field data collection yearly or every 3 years

Site: Water

08 Water quality: Tanzania River Scoring System



Site level

Rapid bioassessment protocols are commonly used to determine ecosystem health, based on the premise that indicator species are sensitive to changes in habitat/water quality from anthropogenic pressures.

Ecosystem services/disservices: freshwater, water regulation, erosion control, water purification

Collection method: Field data collection yearly or every 3 years

09 Depth to water table



Site level

Fluctuations in water table depth indicate across seasons indicate ground water depletion and within seasons indicate how water availability varies throughout the year.

Ecosystem services/disservices: freshwater, water regulation, food and fiber

Collection method: Field data collection yearly or every 3 years

10 Infiltration rate



The rate at which water enters a soil

Ecosystem services/disservices: water quality, water availability, nutrient leaching, flood mitigation

Collection method: Field data collection yearly or every 3 years in the dry season

Landscape: Water

11

Water quality: Tanzania River Scoring System



Landscape level

Rapid bioassessment protocols are commonly used to determine ecosystem health, based on the premise that indicator species are sensitive to changes in habitat/water quality from anthropogenic pressures.

Ecosystem services/disservices: freshwater, water regulation, erosion control, water purification

Collection method: Field data collection yearly or every 3 years

12

Access to water per different use types



Landscape level: Access to water

This will influence the system's ability to cope with stress and shocks and can have outcomes for human health and wellbeing.

Ecosystem services/disservices: health, clean water, water availability

Collection method: Household survey yearly or every 3 years

Landscape: Resilience

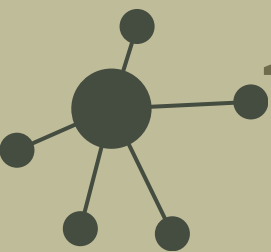


13 Household poverty

Acknowledges and allows exploration of the explicit link between land health and human health.

Ecosystem services/disservices: Education, social relations

Collection method: Household survey yearly or every 3 years

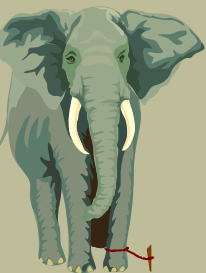


14 Ecosystem Connectivity

Spatial analysis of the fragmentation of natural ecosystem patterns.

Ecosystem services/disservices: Genetic resources,

Collection method: Secondary data of land use and vegetation types



15 Threatened species

Uses IUCN Red List of Threatened Species data to estimate the potential reduction in species extinction risk that could be achieved at a site

Ecosystem services/disservices: Genetic resources, education values, inspiration, recreation and ecotourism

Collection method: Secondary data using the Integrated Biodiversity Tool



16 Productivity

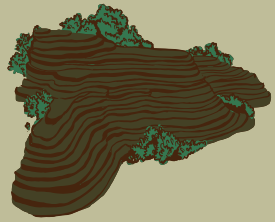
Can link land health directly with economic productivity, food security and sustainable land management practices.

Ecosystem services/disservices: food and fibre, primary productivity

Collection method: Household survey yearly or every 3 years

Landscape: Resilience

17 Sustainable Land Management



Total number and different practices at each site

Ecosystem services/disservices: A driver that can influence different ecosystem services such as food and fibre, freshwater, climate regulation, water regulation, erosion control, pollination, biological control

Collection method: Household survey yearly or every 3 years

18 Perception of shocks and stresses

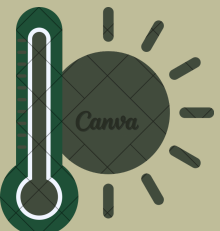


Percentage of those surveyed experiencing different shocks and stresses

Ecosystem services/disservices: Depending on the shock/stress, different ecosystem services may be affected such as food and fibre, regulation of human disease, water regulation, storm protection, freshwater

Collection method: Household survey yearly or every 3 years

19 Extreme weather



No of days at or above 35 degrees celsius and no of days a station records heavy rain of 50 mm or more

Ecosystem services/disservices: yields, food security, soil erosion

Collection method: Secondary data from Tanzania Meteorological Society

Stakeholder	Role in land health
Site	
Farmers	The majority of farmers across both clusters are smallholders. The scale of farming in the clusters means that shifting to sustainable practices could have a very large effect on land health at the landscape level. They influence soil quality at the site level, as well as biodiversity, and water quality and availability.
Agricultural extension officers (AEO)	AEOs are the first port of call for many smallholders seeking farming advice and so are very influential as to how farming is practised at the site level, particularly for pest and input management.
Pastoralists	Through livestock management, pastoralists influence land health on their grazing areas and water quality when livestock use rivers.
Local government authorities (village and ward level):	Through governance of natural resources, local government authorities bridge the gap between site and landscape management.
Landscape	
Water user associations	Water user associations are part of the local authority structure and can be governed as water catchment committees or water user groups that influence local level water availability and quality.
District councils	They manage and oversee natural resources for the district. They enforce and monitor compliance of laws set by the central government for resource use.
Farmer cooperatives	Many farmers growing crops commercially are members which helps farmers market their crops and provides for the local purchase of agricultural input supplies.
Tanzania National Parks Authority	TANAPA is responsible for the management of national parks that are key for land health because they influence many of the LHMI indicators such as connectivity of natural ecosystems, biodiversity, carbon, human-wildlife interactions, habitat for threatened species and will influence water quality and availability due to catchment zones.
Tanzania Forest Services Agency	Manages large areas of national forest reserves influencing many of the LHMI indicators such as connectivity of natural ecosystems, biodiversity, carbon, human-wildlife interactions, habitat for threatened species and will influence water quality and availability due to catchment zones.
Non-governmental organisations	Potential to have a large impact on land health in the localised areas where they work at the site and landscape scale through initiatives such as restoring elephant corridors or managing protected forest. Some organisations such as Southern Tanzania Elephant program have a stronger landscape and national level influence through their work on wildlife corridors. Both clusters: Reforest Africa, Southern Tanzania Elephant Program Ihemi only: Clinton Foundation and partners, Agronomos Sin Fronteras Kilombero only: African Wildlife Foundation
Rufiji Basin Water Authority (RBWA)	RBWA mandates on allocating water resources, controlling water pollution, protecting water sources and other general water resources management matters - therefore they have the ability to influence water use and access across both clusters.
Private companies	Kilombero cluster: Kilombero Sugar Company, Kilombero Valley Teak Company. Ihemi cluster:
National	
SAGCOT centre limited	SAGCOT influences land health by also being active between the landscape and national scale, influencing policy and regulation to meet its objectives.
Government ministries e.g. Ministry of Agriculture, Livestock and Fisheries	They influence land health as stakeholders at the landscape and site level implement their policies within their jurisdiction.
Tanzania Agricultural Research Institute	Their research focuses on how to help farmers improve production such as on crop varieties, fertilisers, and agronomic practices.
National Carbon Monitoring Centre	It is a vehicle for reporting on carbon stocks and their changes as well as coordinating the national measuring, reporting and verification processes for the Government of Tanzania.

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